BE (E&TC)	PLC SCADA and Automation (Elective-III)	
Experiment No: 1	To design and implement basic logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) using PLC programming and verify their operation through simulation (software) and hardware connection.	**Pages**

Aim: Simulate & implement basic logic gates ladder logic program.
Software Used: TIA 13(Totally Integrated Automation)
Theory:

A Programmable Logic Controller, PLC, or Programmable Controller is an electronic device used for Automation of industrial processes, such as control of machinery on factory assembly lines. A programmable controller is a digitally operating electronic apparatus which uses a programmable memory for the internal storage of instructions for implementing specific functions, such as logic, sequencing, timing, counting and arithmetic, to control various machines or processes through digital or analog input/output devices. Unlike general purpose computers, the PLC is designed for multiple inputs and output arrangements, extended temperature ranges, immunity to electrical noise, and resistance to vibrations and impacts.

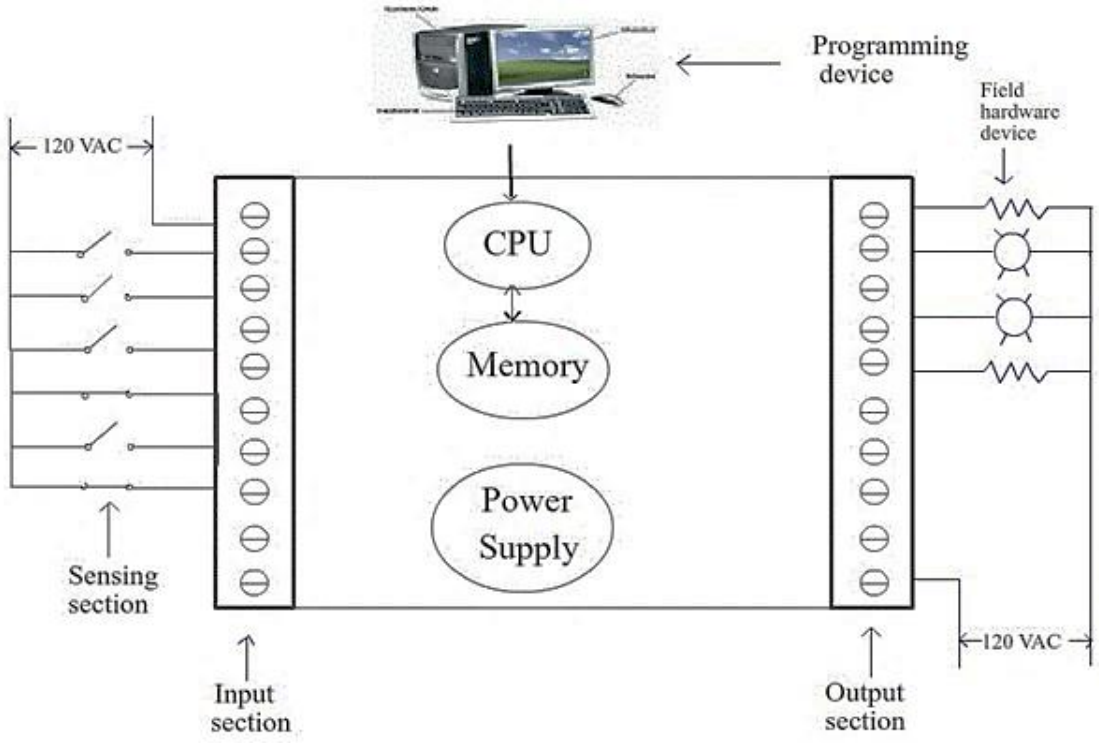


Fig. 1: Block Diagram of a PLC

A PLC has following sections and each section has unique job to perform.

1. **The sensing section:** This section consists of limit switches, photoelectric sensors, push buttons etc. These incoming hardware devices provides input signal to the PLC. These devices are also

called as field input devices. The term “field input " is used because this device provides incoming signals that are tangible items that you physically connect to PLC.

2. **Input section:** This section is majorly divided into 2 parts:
   First, the physical screw terminals, where incoming signal (i.e. input), from the field input devices (e.g. limit switch) are connected to the PLC.
   The second portion of them input section is the PLC's internal conversion electronics. This section converts and isolates the high-voltage input level from field input devices. High-voltage signals from field input devices are converted to +5 volts direct current (VDC) for a valid ON input signal, and a 0 VDC for a valid OFF input signal. Incoming signal conversion and isolation is necessary because microprocessor components operate on +5 VDC, whereas an input signal may be of 24 VDC, 120 volts alternating current (VAC), or 220 VDC. If 120 VAC signal is inputted, for example, into 5 VDC, circuit will quickly destroy your PLC.
3. **Controller:** The controller is also known as central processing unit (CPU), or simply as the processor. Central processing unit controls or supervises the entire process. The central processing unit solves the user program and apparently updates the status of the outputs.
4. **Programmer:** The programmer is a device used by the programmer or operator to enter or edit program instructions or data. The programmer can be handheld unit, a personal computer, or an industrial computer programming terminal.
5. **Output section:** The ON or OFF status of the inputs are read and the information is used to solve the user ladder program and the updated signals is sent to the output section. The output section is simply a series of switches, one for each output point, that are controlled by CPU and are used to turn output field devices ON or OFF.
6. **Field hardware devices:** The devices that are controlled by the PLC's output section screw terminals are the field hardware devices.

Each manufacturer of PLC systems has own style of writing the instructions. Different PLCs has different instruction sets but even some common basic instructions are shared by all the PLCs. All manufacturers give different software packages for programming PLCs. Ladder is most commonly used programming language. Prior to PLCs, relay logic was used in industry. Ladders were developed to mimic or imitate relay logic.

Relay Logic / Instructions:

A relay is simple magnetic device which acts as a control switch. When the switch is on, current will flow through the coil on iron piece. This iron core acts a electromagnet and due to the magnetic field upper contact gets attracted towards lower one and circuit gets completed, allowing current to flow from load.

Ladder Programming:

Ladder diagram is popular language of programming the PLCs. Ladder diagram shows the sequence of the logic execution which is presented diagrammatically. In ladder diagram, there are two vertical lines generally called as Phase (positive) or neutral. Rungs which show current flow in horizontal direction are the sequence in which the logic executes. The Analogous to relay, ladder has two main symbols which are contacts and output coil. Generally each rung has inputs (contacts) on left hand side and outputs (coil) on the right hand side. These contacts and coils are called as bits of the relays. Each input and output are individual bit in I/O files. An instruction in ladder instructs PLCs how to respond to the bits in I/O files which are stored in the memory. Input contacts are the condition area; the conditions must be fulfilled to change the status of the output coils.

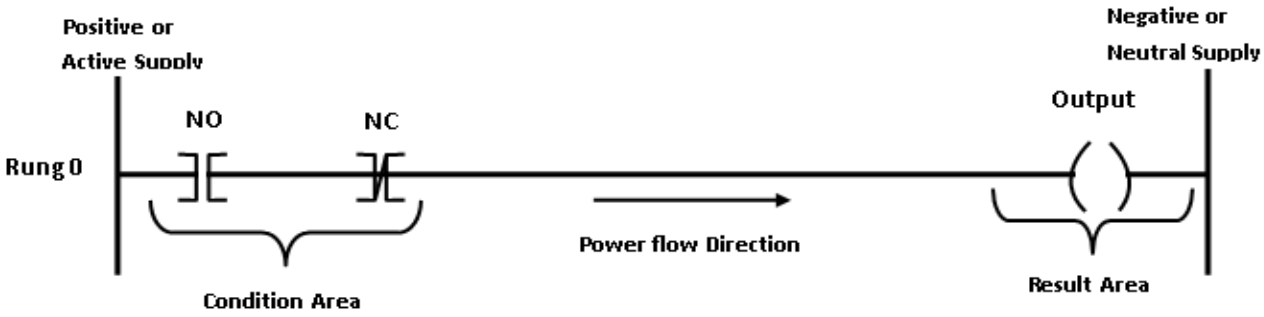
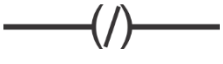


Fig.2: Typical Ladder Diagram

Most commonly used relay instructions used in PLC programming are as shown in the table below:

Instruction	Symbol	Description
Examine ON (Normally Open)		An input condition that is open when de energized
Examine OFF (Normally Closed)		An input condition that is close when de energized
Output Coil		An output instruction that is true when the input conditions become true
Negated Output		An output instruction that is true all the time except when all input conditions true
Latch Output Coil		To hold an output ON
Unlatch		To unlatch the latched ON output

The Binary Concept:

The PLC, like all digital equipment, operates on the binary principle. The term binary principle refers to the idea that many things can be thought of as existing in only one of two

states. These states are 1 and 0. The 1 and 0 can represent ON or OFF, open or closed, true or false, high or low, or any other two conditions.

Logic Function:

A *logic gate* is a circuit with several inputs but only one output that is activated by particular combinations of input conditions. The two-state binary concept, applied to gates, can be the basis for making decisions.

Logic is the ability to make decisions when one or more different factors must be taken into account before an action is taken. This is the basis for the operation of the PLC, where it is required for a device to operate when certain conditions have been met.

The AND Function

An AND gate is a device with two or more inputs and one output. The AND gate output is 1 only if all inputs are 1. If any input is 0, the output will be 0. The AND logic gate operates similarly to control devices connected in series.

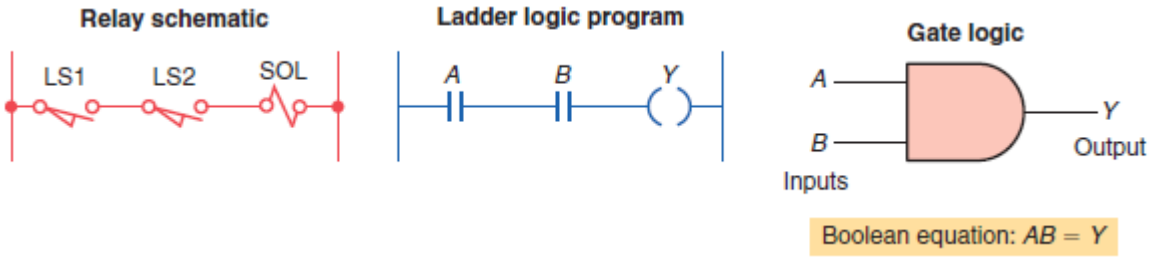


Fig.3 (a): AND Logic

Truth Table for AND gate:

The OR Function

An OR gate can have any number of inputs but only one output. The OR gate output is 1 if one or more inputs are 1. If all inputs are 0, the output will be 0. The OR logic gate operates similarly to control devices connected in parallel.

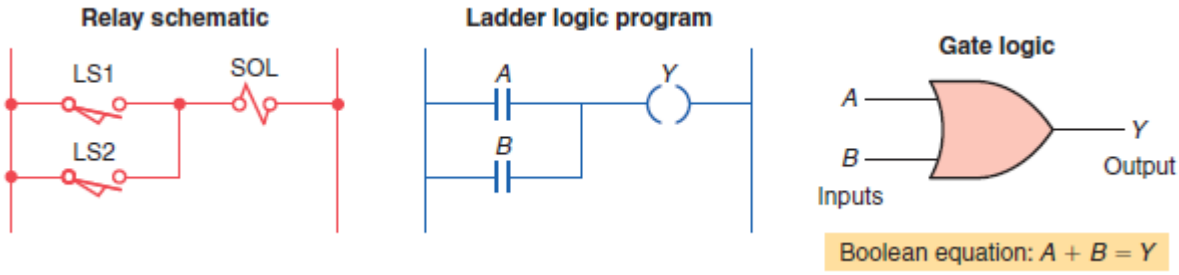


Fig.3 (b): OR Logic

Truth Table for OR gate:

The NOT Function

Unlike the AND and OR functions, the NOT function can have *only one* input. The NOT output is 1 if the input is 0. The output is 0 if the input is 1. The result of the NOT operation is always the inverse of the input, and the NOT function is, therefore, called an *inverter.* The logical NOT function can be performed on a contact input simply by using a normally closed instead of a normally open contact.

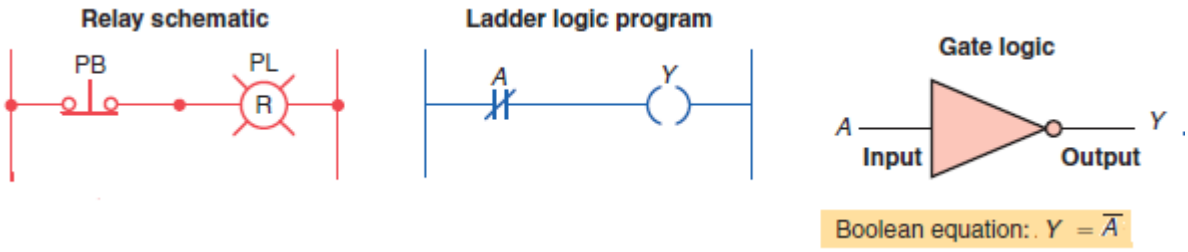


Fig.3 (c): NOT Logic

Truth Table for NOT gate:

The NAND Function

The NOT symbol placed at the output of an AND gate would invert the normal output result.

An AND gate with an inverted output is called a *NAND* gate.

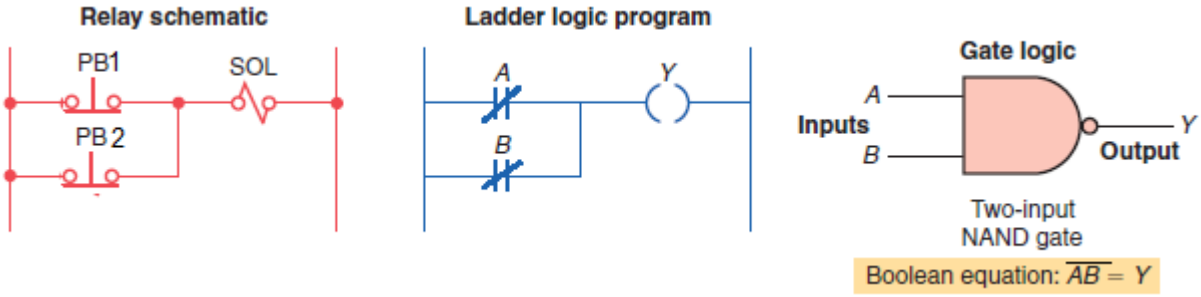


Fig.3 (d): NAND Logic

Truth Table for NAND gate:

The NOR Function

The NOT symbol placed at the output of an OR gate would invert the normal output result. An OR gate with an inverted output is called a *NOR* gate.

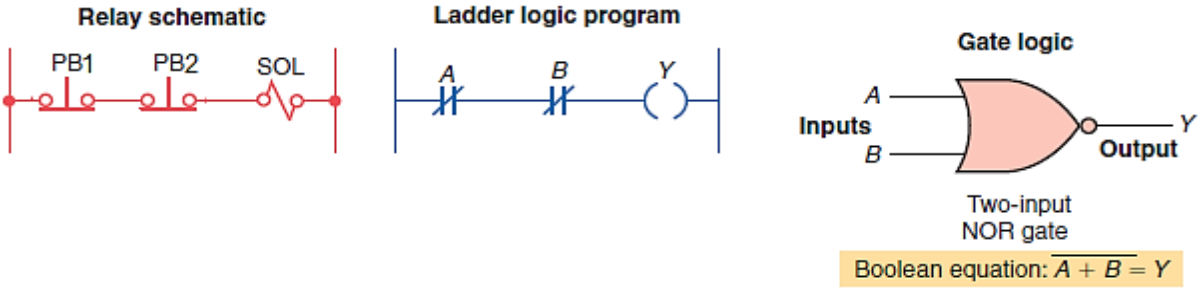


Fig.3 (e): NOR Logic

Truth Table for NOR gate:

The Exclusive-OR (XOR) Function

The output of this *XOR* gate is HIGH only when one input or the other is HIGH, but not both. The exclusive-OR gate is commonly used for the *comparison* of two binary numbers.

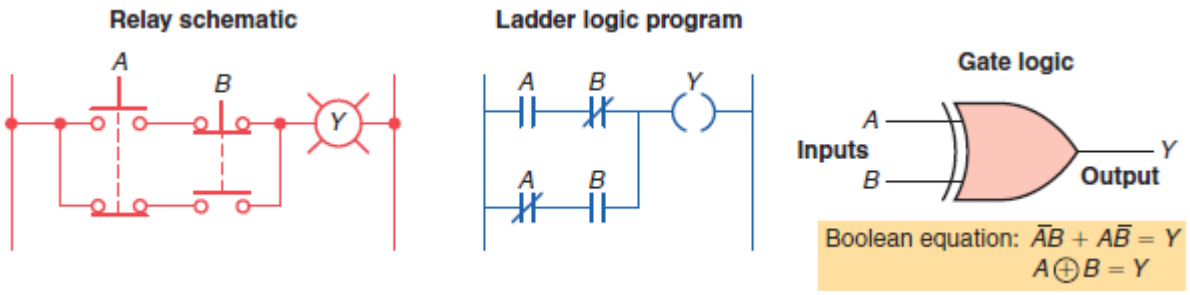


Fig.3 (f): Exclusive-OR (XOR) Logic

Truth Table for gate: Exclusive-OR

Conclusion:

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Questions:

1. Define PLC.
2. Sketch the block diagram of a PLC and explain its parts in detail.
3. Illustrate the functions of an electromechanical Relay?
4. Express each of the following equations as a ladder logic program::

(i) $Y = A\overline{B}$
(ii) $Y = AB + CD$

REFERENCES:

[1]Petruzella, F. D. (2017). *Programmable Logic Controllers* (5th ed.). McGraw-Hill Education.
[2] Bolton, W. (2021). *Programmable Logic Controllers* (7th ed.). Elsevier.
[3] Siemens AG. (2023). *TIA Portal Programming Manual*.Siemens Industry Online Support (SIOS).